CHAPTER 13

Digital Integrated Circuits

13.1 Introduction

The integrated circuit (IC) was introduced in Section 1-9, and the various IC digital logic families were discussed in Section 2-8. This chapter presents the basic electronic circuits in each IC digital logic family and analyzes their electrical operation. A basic knowledge of electronics is assumed.

The IC digital logic families to be considered here are:

RTL	Resistor-transistor logic
DTL	Diode-transistor logic
I^2L	Integrated-injection logic
TTL	Transistor-transistor logic
ECL	Emitter-coupled logic
MOS	Metal-oxide semiconductor
CMOS	Complementary metal-oxide semiconductor

The first two, RTL and DTL, have only historical significance since they are seldom used in new designs. RTL was the first commercial family to have been used extensively. It is included here because it represents a useful starting point for explaining the basic operation of digital gates. DTL circuits have been gradually replaced by TTL. In fact, TTL is a modification of the DTL gate. The operation of the TTL gate will be easier to understand after the DTL gate is discussed. The characteristics of TTL, ECL, and CMOS were presented in Section 2-8. These families have a large number of SSI circuits, as well as MSI and LSI circuits. I^2L and MOS are mostly used for constructing LSI functions.

The basic circuit in each IC digital logic family is either a NAND or a NOR gate. This basic circuit is the primary building block from which more complex functions are obtained. An *RS* latch is constructed from two NAND or two NOR gates connected back to back. A master-slave flip-flop is obtained from the interconnection of about ten basic gates. A register is obtained from the interconnection of flip-flops and basic gates. Each IC logic family has available a catalog of integrated-circuit packages that provide various digital logic functions. The differences in the logic functions available from each logic family are not so much in the function that they achieve as in the specific characteristics of the basic gate from which the function has been constructed.

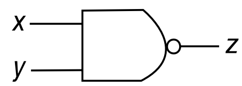
NAND and NOR gates are usually defined by the Boolean functions they implement in terms of binary variables. When analyzing them as electronic circuits, it is more convenient to

Inputs		Output
x	*y*	*z*
L	L	H
L	H	H
H	L	H
H	H	L

NAND gate

(a) If *any* input is LOW, the output is HIGH.

(b) If *all* inputs are HIGH, the output is LOW.



Inputs		Output
x	*y*	*z*
L	L	H
L	H	L
H	L	L
H	H	L

NOR gate

(a) If *any* input is HIGH, the output is LOW.

(b) If *all* inputs are LOW, the output is HIGH

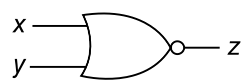


Figure 13.1 Input-output conditions for positive-logic NAND and NOR gates

investigate their input-output relationships in terms of two voltage levels: a *high* level (*H*) and a *low* level (*L*) (see Fig. 2-10). Binary variables take the values 1 and 0. When positive logic is adopted, the high voltage level is assigned the binary value of 1, and the low voltage level a binary 0. From the truth table of a positive-logic NAND gate, we deduce its behavior in terms of high and low levels as stated in Fig. 13-1. The corresponding behavior of the NOR gate is also stated in the same figure. These statements must be remembered because they will be used during the analysis of all the gates in this chapter.

The various digital logic families are usually evaluated by comparing the characteristics of the basic gate in each family. The most important characteristics were discussed in Section 2-8. They are listed here again for reference.

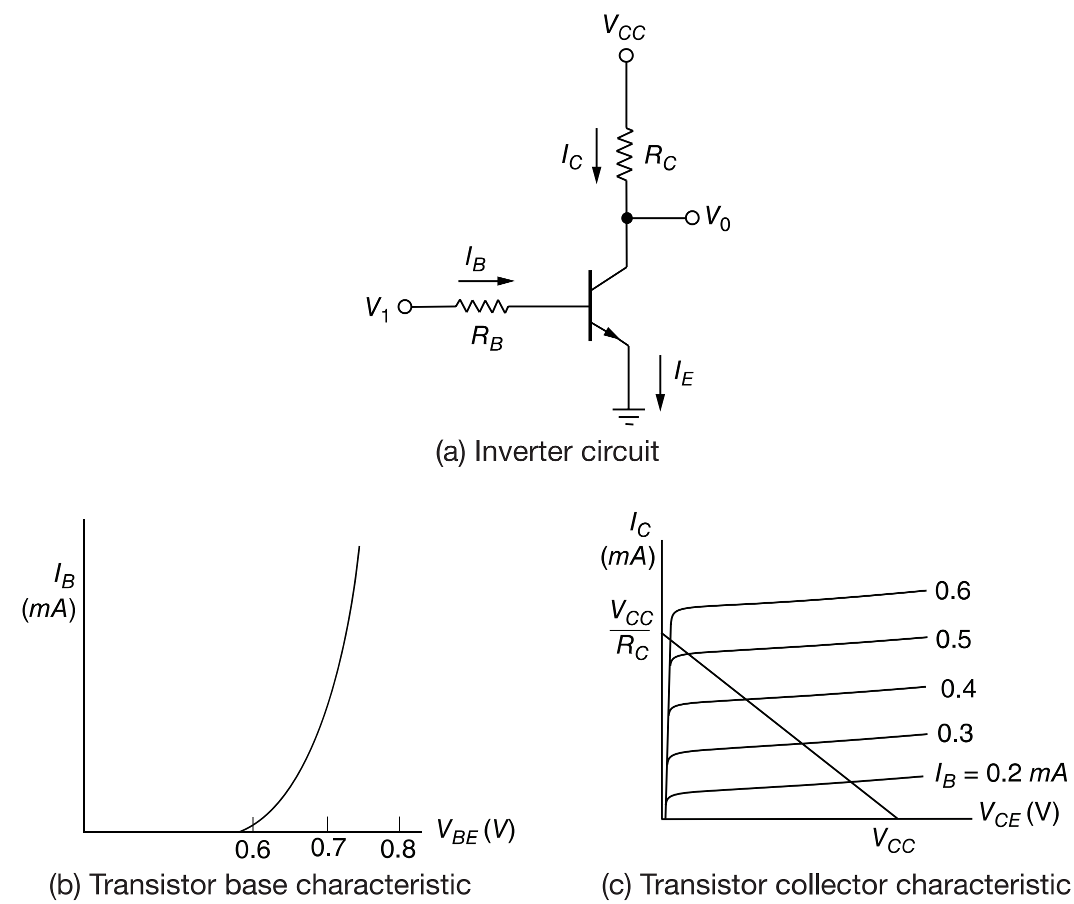
1. *Fan-out* specifies the number of standard loads that the output of the gate can drive without impairment of its normal operation. A standard load is defined as the current flowing in the input of a gate in the same IC family.
2. *Power dissipation* is the power consumed by the gate, which must be available from the power supply.
3. *Propagation delay* is the average transition delay time for the signal to propagate from input to output when the signals change in value.
4. *Noise margin* is the limit of a noise voltage which may be present without impairing the proper operation of the circuit.

The *bipolar* junction transistor (BJT) is the familiar *npn* or *pnp* junction transistor. In contrast, the field-effect transistor (FET) is said to be *unipolar*. The operation of a bipolar transistor depends on the flow of two types of earners: electrons and holes. A unipolar transistor depends on the flow of only one type of majority carrier which may be eletrons (n-channel) or holes (p-channel). The first five logic families listed previously, RTL, DTL, TTL, ECL, and I^2L, use bipolar transistors. The last two logic families, MOS and CMOS, employ a type of unipolar transistor called metal-oxide semiconductor field-effect, transistor, abbreviated MOSFET, or MOS for short. We begin by describing the characteristics of the bipolar transistor and the basic gates used in the bipolar logic families. We then explain the operation of the MOS transistor in conjunction with its two logic families.

13.2 Bipolar Transistor Characteristics

This section is devoted to a review of the bipolar transistor as applied to digital circuits. This information will be used for the analysis of the basic circuit in the five bipolar logic families. Bipolar transistors may be of the *npn* or *pnp* type. Moreover, they are constructed either with germanium or silicon semiconductor material. IC transistors, however, are made with silicon and are usually of the npn type.

The basic data needed for the analysis of digital circuits may be obtained from inspection of the typical characteristic curves of a common-emitter *npn* silicon transistor, shown in Fig. 13-2. The circuit in (a) is a simple inverter with two resistors and a transistor. The current marked I_C flows through resistor R_C and the collector of the transistor. Current I_B flows through resistor R_B and the base of the transistor. The emitter is connected to ground and its current $I_E = I_C + I_B$. The supply voltage is between V_{CC} and ground. The input is between V_i and ground, and the output is between V_0 and ground.



(a) Inverter circuit

(b) Transistor base characteristic

(c) Transistor collector characteristic

Figure 13.2 Silicon *npn* transistor characteristics

We have assumed a positive direction for the currents as indicated. These are the directions in which the currents normally flow in an *npn* transistor. Collector and base currents I_C and I_B are positive when they flow into the transistor. Emitter current I_E is positive when it flows out of the transistor, as indicated by the arrow in the emitter terminal. The symbol V_{CE} stands for the voltage drop from collector to emitter and is always positive. Correspondingly, V_{BE} is the voltage drop across the base-to-emitter junction. This junction is forward biased when V_{BE} is positive. It is reverse biased when V_{BE} is negative.

The base-emitter graphical characteristic is shown in Fig. 13-2(b). This is a plot of V_{BE} versus I_B. If the base-emitter voltage is less than 0.6 V, the transistor is said to be *cutoff* and no base current flows. When the base-emitter junction is forward biased with a voltage greater than 0.6 V, the transistor conducts and I_B starts rising very fast while V_{BE} changes very little. The voltage V_{BE} across a conducting transistor seldom exceeds 0.8 V.

The graphical collector-emitter characteristics, together with the load line, are shown in Fig. 13-2(c). When V_{BE} is less than 0.6 V, the transistor is cutoff with $I_B = 0$ and a negligible current flows in the collector. The collector-to-emitter circuit then behaves like an open circuit. In the *active* region, collector voltage V_{CE} may be anywhere from about 0.8 V up to V_{CC}. Collector current I_C in this region can be calculated to be approximately equal to $I_B h_{FE}$, where h_{FE} is a transistor parameter called the *dc current gain.* The maximum collector current depends not on I_B, but rather on the external circuit connected to the collector. This is because V_{CE} is always positive and its lowest possible value is 0 V. For example, in the inverter shown, the maximum I_C is obtained by making $V_{CE} = 0$ to obtain $I_C = V_{CC}/R_C$.

It was stated that $I_C = h_{FE} I_B$ in the active region. The parameter h_{FE} varies widely over the operating range of the transistor, but still it is useful to employ an average value for the purpose of analysis. In a typical operating range, h_{FE} is about 50, but under certain conditions it could go down to as low as 20. It must be realized that the base current I_B may be increased to any desirable value, but the collector current I_C is limited by external circuit parameters. As a consequence, a situation can be reached where $h_{FE} I_B$ is greater than I_C. If this condition exists, then the transistor is said to be in the *saturation* region. Thus, the condition for saturation is determined from the relationship:

$$I_B \geq \frac{I_{CS}}{h_{FE}}$$

where I_{CS} is the maximum collector current flowing during saturation. V_{CE} is not exactly zero in the saturation region but is normally about 0.2V.

The basic data needed for analyzing bipolar transistor digital circuits are listed in Table 13-1. In the cutoff region, V_{BE} is less than 0.6 V, V_{CE} is considered as an open circuit, and both

Table 13-1 Typical npn silicon transistor parameters

Region	V_{BE}(V)*	V_{CE}(V)	Current relationship
Cutoff	< 0.6	Open circuit	$I_B = I_C = 0$
Active	0.6 – 0.7	> 0.8	$I_C = h_{FE} I_B$
Saturation	0.7 – 0.8	0.2	$I_B \geq I_{CS}/h_{FE}$

* V_{BE} will be assumed to be 0.7 V if the transistor is conducting, whether in the active or saturation region.

currents are negligible. In the active region, V_{BE} is about 0.7 V, V_{CE} may vary over a wide range, and I_C can be calculated as a function of I_B. In the saturation region. V_{BE} hardly changes but V_{CE} drops to 0.2 V. The base current must be large enough to satisfy the inequality listed. To simplify the analysis, we will assume that $V_{BE} = 0.7$ V if the transistor is conducting, whether in the active or saturation region.

The analysis of digital circuits may be undertaken using a prescribed procedure: For each transistor in the circuit determine if its V_{BE} is less than 0.6 V. If so, then the transistor is cutoff and the collector-to-emitter circuit is considered an open circuit. If V_{BE} is greater than 0.6 V, the transistor may be in the active or saturation region. Calculate the base current, assuming that $V_{BE} = 0.7$ V. Then calculate the maximum possible value of collector current I_{CS}, assuming $V_{CE} = 0.2$ V. These calculations will be in terms of voltages applied and resistor values. Then, if the base current is large enough that $I_B > I_{CS}/H_{FE}$, we deduce that the transistor is in the saturation region with $V_{CE} = 0.2$ V. However, if the base current is smaller and the above relationship is not satisfied, the transistor is in the active region and we recalculate collector current I_C using the equation $I_C = h_{FE}I_B$.

To demonstrate with an example, consider the inverter circuit of Fig. 13-2(a) with the following parameters:

$R_C = 1$ kΩ	$V_{CC} = 5$ V (voltage supply)
$R_B = 22$ kΩ	$H = 5$ V (high-level voltage)
$h_{FE} = 50$	$L = 0.2$ V (low-level voltage)

With input voltage $V_i = L = 0.2$ V, we have that $V_{BE} < 0.6$ V and the transistor is cutoff. The collector-emitter circuit behaves like an open circuit; so output voltage $V_0 = 5\text{ V} = H$.

With input voltage $V_i = H = 5$ V, we deduce that $V_{BE} > 0.6$ V. Assuming that $V_{BE} = 0.7$, we calculate the base current:

$$I_B = \frac{V_i - V_{BE}}{R_B} = \frac{5-07}{22K\Omega} = 0.195\ mA$$

The maximum collector current, assuming $V_{CE} = 0.2$ V, is:

$$I_{CS} = \frac{V_{CC} - V_{CE}}{R_C} = \frac{5-0.2}{1k\Omega} = 4.8\ mA$$

We then check for saturation:

$$0.195 = I_B \geq \frac{I_{CS}}{h_{FE}} = \frac{4.8}{50} = 0.096\ mA$$

and find that the inequality is satisfied since 0.195 > 0.096. We conclude that the transistor is saturated and output voltage $V_0 = V_{CE} = 0.2\ V = L$. Thus the circuit behaves as an inverter.

The procedure just described will be used extensively during the analysis of the circuits in the following sections. This will be done by means of a qualitative analysis, i.e., without writing

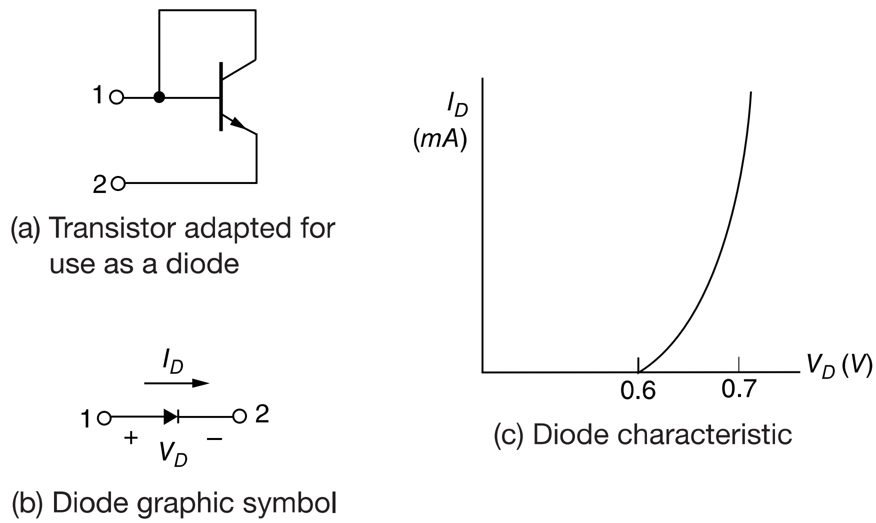


Figure 13.3 Silicon diode symbol and characteristic

down the specific numerical equations. The quantitative analysis and specific calculations will be left as exercises in the Problems section at the end of the chapter.

There are occasions where not only transistors but also diodes are used in digital circuits. An IC diode is usually constructed from a transistor with its collector connected to the base, as shown in Fig, 13-3(a). The graphic symbol employed for a diode is shown in Fig. 13-3(b). The diode behaves essentially like the base-emitter junction of a transistor. Its graphical characteristic, shown in Fig. 13-3(c), is similar to the base-emitter characteristic of a transistor. We can then conclude that a diode is off and nonconducting when its forward voltage, V_D, is less than 0.6 V. When the diode conducts, current I_D flows in the direction shown in Fig. 13-3(b), and V_D stays at about 0.7 V. One must always provide an external resistor to limit the current in a conducting diode, since its voltage remains fairly constant at a fraction of a volt.

13.3 RTL and DTL Circuits

13.3.1 RTL Basic Gate

The basic circuit of the RTL digital logic family is the NOR gate shown in Fig, 13-4. Each input is associated with one resistor and one transistor. The collectors of the transistors are tied together at the output. The voltage levels for the circuit are 0.2 V for the low-level and from 1 to 3.6 V for the high-level.

The analysis of the RTL gate is very simple and follows the procedure outlined in the previous section. If any input of the RTL gate is high, the corresponding transistor is driven into saturation. This causes the output to be low, regardless of the states of the other transistors. If all inputs are low at 0.2 V, all transistors are cutoff because $V_{BE} < 0.6$ V. This causes the output of the circuit to be high, approaching the value of supply voltage V_{CC}. This confirms the conditions stated in Fig. 13-1 for the NOR gate. Note that the noise margin for low signal input is 0.6 − 0.2 = 0.4 V.

The fan-out of the RTL gate is limited by the value of the output voltage when high. As the output is loaded with inputs of other gates, more current is consumed by the load. This current must flow through the 640-Ω resistor. A simple calculation (see Problem 13-1) will show that, if

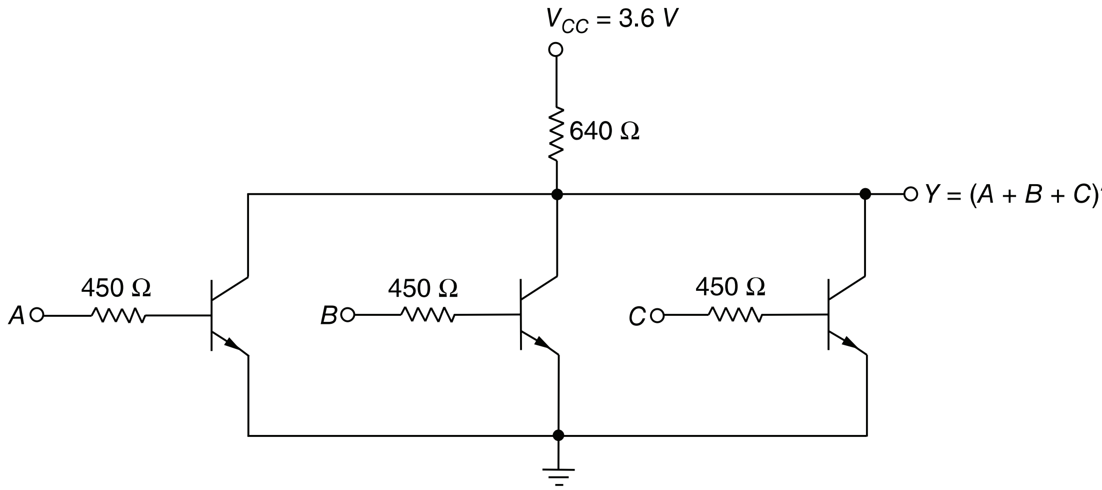


Figure 13.4 RTL basic NOR gate

h_{FE} drops to 20, the output voltage drops to about 1 V when the fan-out is 5. Any voltage below 1 V in the output may not drive the next transistor into saturation as required. The power dissipation of the RTL gate is about 12 mW and the propagation delay averages 25 ns.

13.3.2 DTL Basic Gates

The basic circuit in the DTL digital logic family is the NAND gate shown in Fig. 13-5. Each input is associated with one diode. The diodes and the 5-$k\Omega$ resistor form an AND gate. The transistor serves as a current amplifier while inverting the digital signal. The two voltage levels are 0.2 V for the low-level and between 4 and 5 V for the high-level.

The analysis of the DTL gate should conform to the conditions listed in Fig. 13-1 for the NAND gate. If any input of the gate is low at 0.2 V, the corresponding input diode conducts current through V_{CC} and the 5-kΩ resistor into the input node. The voltage at point P is equal to the input voltage of 0.2 V plus a diode drop of 0.7 V, for a total of 0.9 V. In order for the transistor

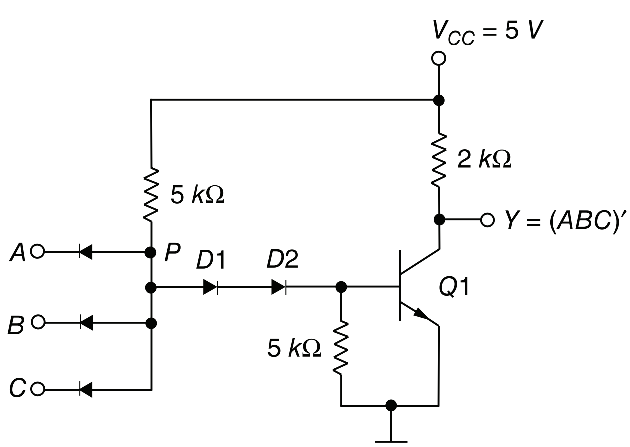


Figure 13.5 DTL basic NAND gate

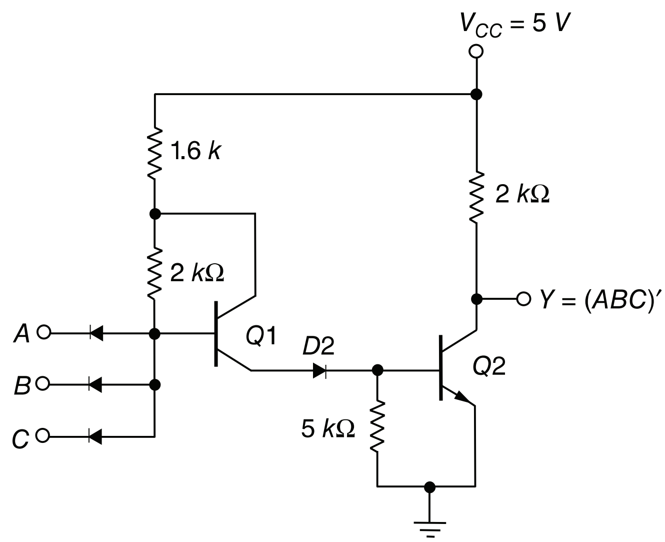


Figure 13.6 Modified DTL gate

to start conducting, the voltage at point P must overcome a potential of one V_{BE} drop in Q 1 plus two diode drops across $D1$ and $D2$, or $3 \times 0.6 = 1.8$ V. Since the voltage at P is maintained at 0.9 V by the input conducting diode, the transistor is cutoff and the output voltage is high at 5 V.

If all inputs of the gate are high, the transistor is driven into the saturation region. The voltage at P now is equal to V_{BE} plus the two diode drops across $D1$ and $D2$, or $0.7 \times 3 = 2.1$ V. Since all inputs are high at 5 V and $V_P = 2.1$ V, the input diodes are reverse biased and off. The base current is equal to the difference of currents flowing in the two 5-kΩ resistors and is sufficient to drive the transistor into saturation (see Problem 13-2). With the transistor saturated, the output drops to V_{CE} of 0.2 V. which is the low level for the gate.

The power dissipation of a DTL gate is about 12 mW and the propagation delay averages 30 ns. The noise margin is about 1 V and a fan-out as high as 8 is possible. The fan-out of the DTL gate is limited by the maximum current that can flow in the collector of the saturated transistor (see Problem 13-3).

The fan-out of a DTL gate may be increased by replacing one of the diodes in the base circuit with a transistor as shown in Fig. 13-6. Transistor $Q1$ is maintained in the active region when output transistor $Q2$ is saturated. As a consequence, the modified circuit can supply a larger amount of base current to the output transistor. The output transistor can now draw a larger amount of collector current before it goes out of saturation. Part of the collector current comes from the conducting diodes in the loading gates when $Q2$ is saturated. Thus, an increase in allowable collector saturated current allows more loads to be connected to the output, which increases the fan-out capability of the gate.

13.3.3 High-Threshold Logic—HTL

There are occasions where digital circuits must operate in an environment which produces very high noise signals. For operation in such surroundings, there is available a type of DTL gate which possesses a high threshold to noise immunity. This type of gate is called a *high-threshold-logic* (HTL) gate.

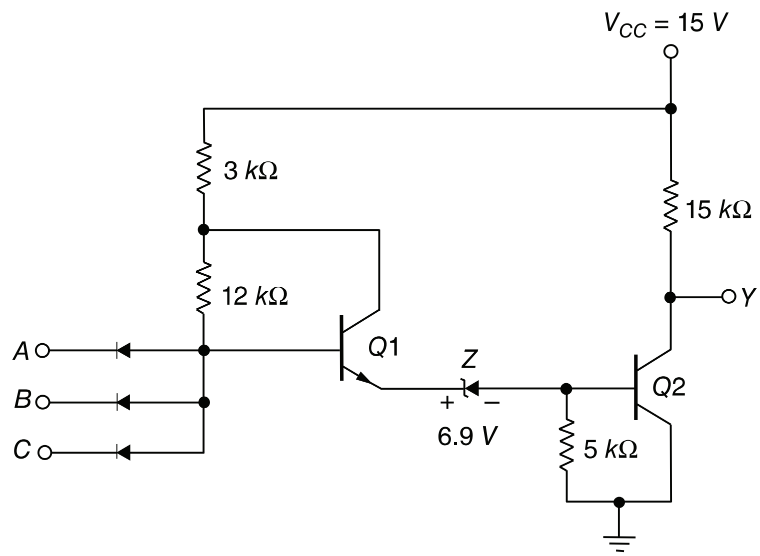


Figure 13.7 High-threshold-logic (HTL) gate

The HTL gate is shown in Fig. 13-7. Comparing it with the modified DTL gate of Fig. 13-6, we note that the supply voltage has been raised to 15 V and a zener diode (Z) is used instead of a normal diode. The zener diode has the characteristic of maintaining a constant voltage of 6.9 V when reverse biased.

In order for output transistor *Q*2 to conduct, the emitter of *Q*1 must rise to a potential of one V_{BE} drop plus the fixed zener voltage of 6.9 V, for a total of about 7.5 V, The low level for the gate remains at 0.2 V, but the high level is about 15 V. With the input of 0.2 V, the base of *Q*1 is at 0.9 V and *Q*2 is off. The noise signal must be greater than 7.5 V to change the state of *Q*2. With all the inputs at 15 V, output transistor *Q*2 is saturated. The noise signal must be greater than 7.5 V (in the negative direction) to turn the transistor off. Thus, the noise margin of the HTL gate is about 7.5 V for both voltage levels.

13.4 Integrated-injection Logic (I^2L)

Integrated-injection logic is the most recent digital logic family to be introduced commercially. Its main advantage is the high packing density of gates that can be achieved in a given area of semiconductor chip. This allows more circuits to be placed in the chip to form complex digital functions. As a consequence, this family is used mostly for LSI functions. It is not available in SSI packages containing individual gates.

The I^2L basic gate is similar in operation to the RTL gate, with few major differences: (1) The base resistor used in the RTL gate is removed altogether in the I^2L gate. (2) The collector resistor used in the RTL gate is replaced by a pnp transistor that acts as a load for the I^2L gate. (3) I^2L transistors use multiple collectors instead of the individual transistors employed in RTL.

The schematic diagram of the basic I^2L gate is shown in Fig. 13-8. It has an npn transistor, *Q*1, with multiple collectors for the outputs. The base circuit has a pnp transistor, *T*1, connected to supply voltage V_{BB}. Unlike other logic families, the I^2L basic gate operation cannot be analyzed when standing alone. One must show its interconnection to other gates to make any sense.

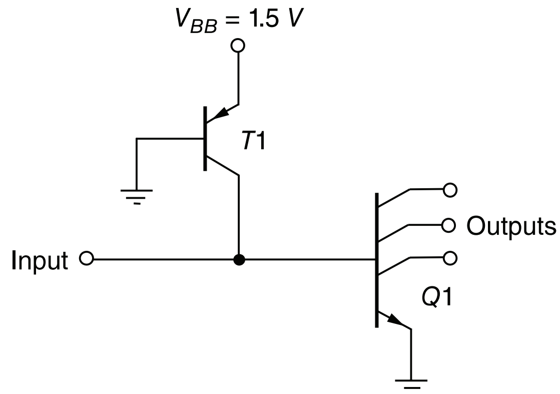
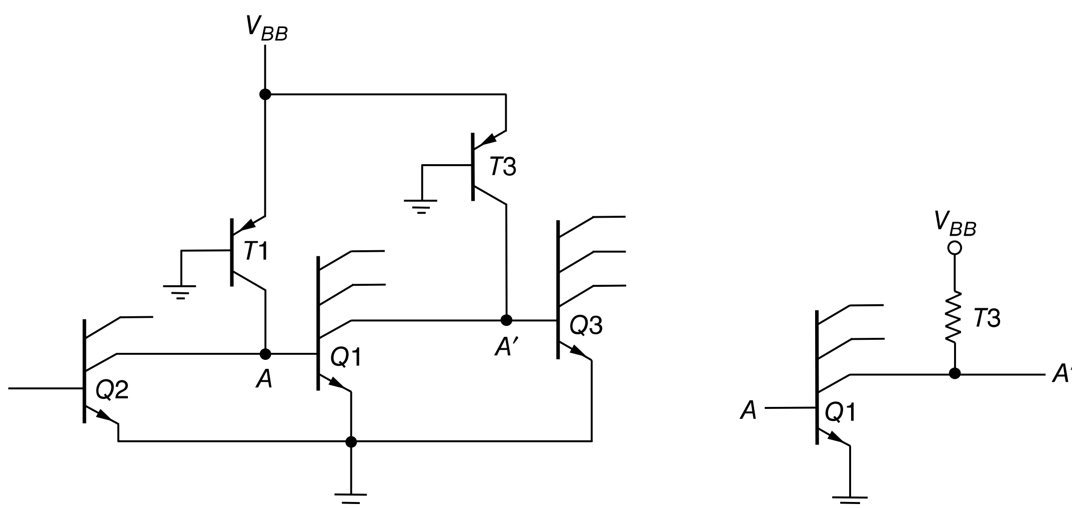


Figure 13.8 I²L basic gate

Figure 13-9 shows the interaction of the basic gate formed by $Q1$ and $T1$ with other gates in its input and output. Here we see that one collector of $Q2$ supplies the input to the basic gate. Transistor $T1$ in the basic gate acts as a load that injects current to the collector of $Q2$. One of the collectors of $Q1$ acts as an output of the basic gate and is connected to the base of $Q3$. Transistor $T3$, connected to the base of $Q3$, acts as a load to inject current to the collector of $Q1$ in the basic gate. The basic gate here acts as an inverter and its equivalent circuit is shown in Fig. 13-9(b). Using multiple collectors and a pnp transistor instead of a load resistor turns out to be a more efficient method of construction, since they reduce the chip area required and allow the packing of more circuits. The pnp transistor, although shown to be connected to the base of a given gate, acts as a collector load for all the other gates that are connected to this base.

The basic I²L gate, when connected to other gates, performs the NOR logic function. This is demonstrated in the circuit diagram shown in Fig. 13-10. The logic function that the circuit implements is drawn with graphic gate symbols in Fig. 13-10(a), which shows the interconnection of two NOR gates and an inverter. This is implemented with three I²L gates, $Q1$, $Q2$, and $Q3$, as shown in Fig. 13-10(b). The output transistors are also shown for completeness. The collectors



(a) Inverter gate $Q1$ (b) Equivalent circuit

Figure 13.9 Connection of other gates to the inputs and outputs of a basic I²L gate

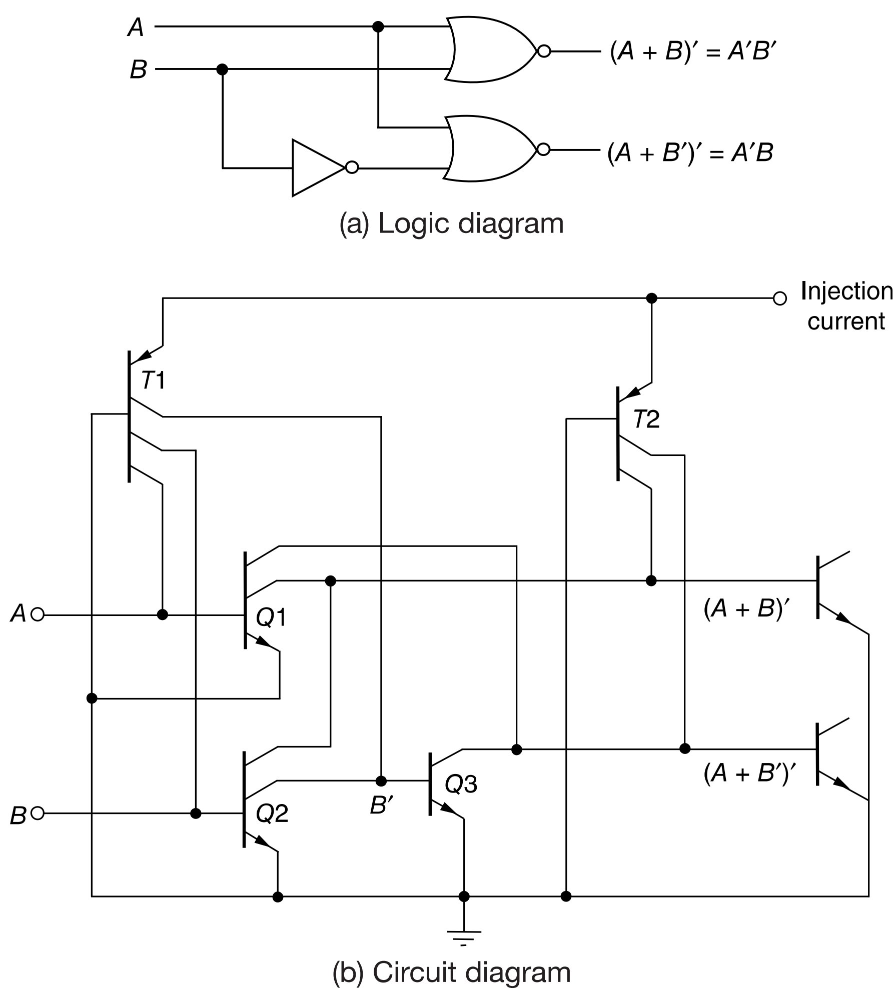


Figure 13.10 Typical connections among I²L gates

of $Q1$ and $Q2$ are tied together to form one NOR function. Input B is complemented by transistor $Q2$. The collectors of $Q3$ and $Q1$ are tied together to form the second NOR function. The base of each npn transistor receives the injection current from the multiple-collector pnp transistors $T1$ and $T2$. The emitters of the npn transistors are connected to the base of the pnp transistor to facilitate the construction.

13.5 Transistor-Transistor Logic (TTL)

The original basic TTL gate was a slight improvement over the DTL gate. As the TTL technology progressed, additional improvements were added to the point where this logic family became the most widely used type in the design of digital systems. There are many versions (or "series") of the TTL basic gate. The names and characteristics of five versions appear in Table 13-2, together with their propagation delay and power dissipation values. The speed-power product is an important parameter for comparing the basic gates. This is a product of the propagation delay and the power dissipation measured in picojoules (pJ). A low value for this parameter is a desirable figure, because it indicates that a given propagation delay can be achieved without excessive power dissipation, or vice versa.

Table 13-2 TTL versions and their characteristics

Name	Abbreviation	Propagation delay (ns)	Power dissipation (mW)	Speed-power product (pJ)
Standard TTL	TTL	10	10	100
Low-power TTL	LTTL	33	1	33
High-speed TTL	HTTL	6	22	132
Schottky TTL	STTL	3	19	57
Low-power Schottky TTL	LSTTL	9.5	2	19

The standard TTL gate was the first version in the TTL family. This basic gate was then constructed with different resistor values to produce gates with lower dissipation or higher speed. The propagation delay of a saturated logic family depends largely on two factors: storage time and *RC* time constants. Reducing the storage time decreases the propagation delay. Reducing resistor values in the circuit reduces the *RC* time constants and decreases the propagation delay. Of course, the trade-off is a higher power dissipation because lower resistances draw more current from the power supply. The speed of the gate is inversely proportional to the propagation delay.

In the low-power TTL gate the resistor values are higher than in the standard gate to reduce the power dissipation, but the propagation delay is increased. In the high-speed TTL gate, resistor values are lowered to reduce the propagation delay, but the power dissipation is increased. The Schottky TTL is a later improvement in the technology that removes the storage time of transistors by preventing them from going into saturation. This version increases the speed of operation without an excessive increase in power dissipation. The low-power Schottky TTL version sacrifices some speed for reduced power dissipation. It is about equal to standard TTL in propagation delay but has only one-fifth the power dissipation. It has the best speed-power product and, as a consequence, it became the most popular version in new designs.

All TTL versions are available in SSI packages and in more complex forms as MSI and LSI functions. The differences in the TTL versions are not in the digital functions that they perform, but rather in the values of resistors and type of transistor that their basic gate uses. In any case, TTL gates in all versions come in three different types of output configurations.

1. Open-collector output.
2. Totem-pole output.
3. Three-state (or tri-state) output.

These three types of outputs will be considered in conjunction with the circuit description of the basic TTL gate.

13.5.1 Open-collector Output Gate

The basic TTL gate shown in Fig. 13-11 is a modified circuit of the DTL gate. The multiple emitters in transistor *Q*1 are connected to the inputs. These emitters behave most of the time like the input diodes in the DTL gate since they form a *pn* junction with their common base. The base-collector junction of *Q*1 acts as another *pn* junction diode corresponding to *D*1 in the DTL gate

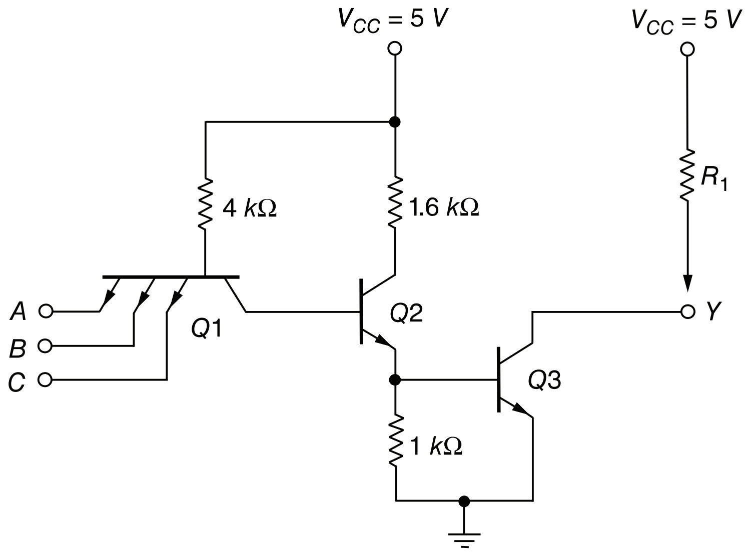


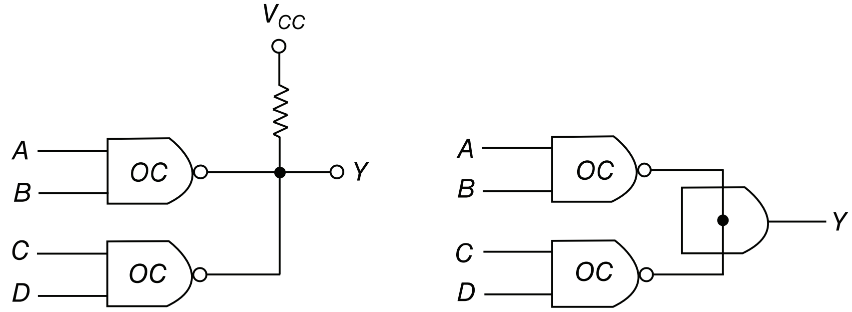
Figure 13.11 Open-collector TTL gate

(see Fig. 13-5). Transistor $Q2$ replaces the second diode, $D2$, in the DTL gate. The output of the TTL gate is taken from the open collector of $Q3$. A resistor connected to V_{CC} must be inserted external to the IC package for the output to "pull up" to the high voltage level when $Q3$ is off; otherwise, the output acts as an open circuit. The reason for not providing the resistor internally will be discussed later.

The two voltage levels of the TTL gate are 0.2 V for the low level and from 2.4 to 5 V for the high level. The basic circuit is a NAND gate. If any input is low, the corresponding base-emitter junction in $Q1$ is forward biased. The voltage at the base of $Q1$ is equal to the input voltage of 0.2 V plus a V_{BE} drop of 0.7 V or 0.9 V. In order for $Q3$ to start conducting, the path from $Q1$ to $Q3$ must overcome a potential of one diode drop in the base-collector *pn* junction of $Q1$ and two V_{BE} drops in $Q2$ and $Q3$, or $3 \times 0.6 = 1.8$ V, Since the base of $Q1$ is maintained at 0.9 V by the input signal, the output transistor cannot conduct and is cutoff. The output level will be high if an external resistor is connected between the output and V_{CC} (or an open circuit if a resistor is not used).

If all inputs are high, both $Q2$ and $Q3$ conduct and saturate. The base voltage of $Q1$ is equal to the voltage across its base-collector *pn* junction plus two V_{BE} drops in $Q2$ and $Q3$, or about $0.7 \times 3 = 2.1$ V. Since all inputs are high and greater than 2.4 V, the base-emitter junctions of $Q1$ are all reverse biased. When output transistor $Q3$ saturates (provided it has a current path), the output voltage goes low to 0.2 V. This confirms the conditions of a NAND operation.

In the above analysis, we said that the base-collector junction of $Q1$ acts like a *pn* diode junction. This is true in the steady-state condition. However, during the turn-off transition, $Q1$ does exhibit transistor action resulting in a reduction in propagation delay. When all inputs are high and then one of the inputs is brought to a low level, both $Q2$ and $Q3$ start turning off. At this time, the collector junction of $Q1$ is reverse biased and the emitter is forward biased; so transistor $Q1$ goes momentarily into the active region. The collector current of $Q1$ comes from the base of $Q2$ and quickly removes the excess charge stored in $Q2$ during its previous saturation state. This causes a reduction in the storage time of the circuit as compared to the DTL type of input. The result is a reduction of the turn-off time of the gate.



(a) Physical connection (b) Wired logic graphic symbol

Figure 13.12 Wired-AND of two open-collector (oc) gates, $Y = (AB + CD)'$

The open-collector TTL gate will operate without the external resistor when connected to inputs of other TIT gates, although this is not recommended because of the low noise immunity encountered. Without an external resistor, the output of the gate will be an open circuit when $Q3$ is off. An open circuit to an input of a TTL gate behaves as if it has a high-level input (but a small amount of noise can change this to a low level). When $Q3$ conducts, us collector will have a current path supplied by the input of the loading gate through V_{CC}, the 4-kΩ resistor, and the forward-biased base-emitter junction.

Open-collector gates are used in three major applications: driving a lamp or relay, performing wired logic, and for the construction of a common-bus system. An open-collector output can drive a lamp placed in its output through a limiting resistor. When the output is low, the saturated transistor $Q3$ forms a path for the current that turns the lamp on. When the output transistor is off, the lamp turns off because there is no path for the current.

If the outputs of several open-collector TTL gates are tied together with a single external resistor, a wired-AND logic is performed. Remember that a positive-logic AND function gives a high level only if all variables are high; otherwise, the function is low. With outputs of open-collector gates connected, together, the common output is high only when all output transistors are off (or high). If an output transistor conducts, it forces the output to the low state.

The wired logic performed with open-collector TTL gates is depicted in Fig. 13-12. The physical wiring in (a) shows how the outputs must be connected to a common resistor. The graphic symbol for such a connection is demonstrated in (b). The AND function formed by connecting together the two outputs is called a wired-AND function. The AND gate is drawn with the lines going through the center of the gate to distinguish it from a conventional gate. The wired-AND gate is not a physical gate but only a symbol to designate the function obtained from the indicated connection. The Boolean function obtained from the circuit of Fig. 13-12 is the AND operation between the outputs of the two NAND gates:

$$Y = (AB)' \cdot (CD)' = (AB + CD)'$$

The second expression is preferred since it shows an operation commonly referred to as an AND-OR-INVERT function (see Section 3-7).

Open-collector gates can be tied together to form a common bus. At any time, all gate outputs tied to the bus, except one, must be maintained in their high state. The selected gate may

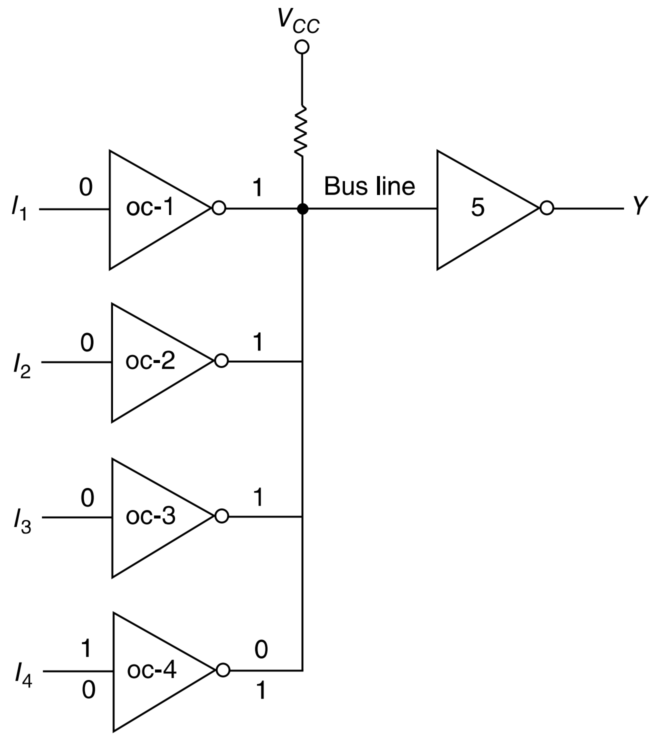


Figure 13.13 Open-collector gates forming a common bus line

be either in the high or low state, depending on whether we want to transmit a 1 or 0 on the bus. Control circuits must be used to select the particular gate that drives the bus at any given time.

Figure 13-13 demonstrates the connection of four sources tied to a common bus line. Each of the four inputs drives an open-collector inverter, and the outputs of the inverters are tied together to form a single bus line. The figure shows that three of the inputs are 0, which produces a 1 or high level on the bus. The fourth input, I_4, can now transmit information through the common bus line into inverter 5. Remember that an AND operation is performed in the wired logic. If $I_4 = 1$, the output of gate 4 is 0 and the wired-AND operation produces a 0. If $I_4 = 0$, the output of gate 4 is 1 and the wired-AND operation produces a 1. Thus, if all other outputs are maintained at 1, the selected gate can transmit its value through the bus. The value transmitted is the complement of I_4, but inverter 5 in the receiving end can easily invert this signal again to make $Y = I_4$.

13.5.2 Totem-pole Output

The output impedance of a gate is normally a resistive plus a capacitive load. The capacitive load consists of the capacitance of the output transistor, the capacitance of the fan-out gates, and any stray wiring capacitance. When the output changes from the low to the high state, the output transistor of the gate goes from saturation to cutoff and the total load capacitance, C, charges exponentially from the low to the high voltage level with a time constant equal to RC. For the open-collector gate, R is the external resistor marked R_L. For a typical operating value of C = 15 pF and R_L = 4 kΩ, the propagation delay of a TTL open-collector gate during the turn-off time is 35 ns. With an active pull-up circuit replacing the passive pull-up resistor R_L, the propagation

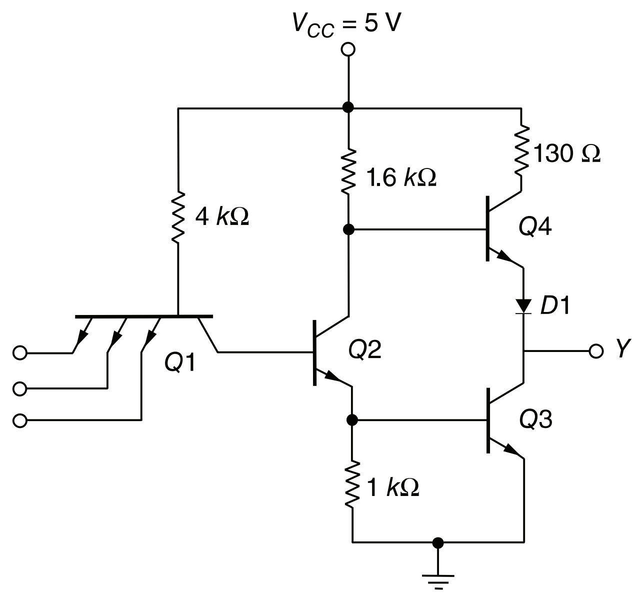


Figure 13.14 TTL gate with totem-pole output

delay is reduced to 10 ns. This configuration, shown in Fig. 13-14, is called a *totem-pole* output because transistor $Q4$ "sits" upon $Q3$.

The TTL gate with the totem-pole output is the same as the open-collector gate, except for the output transistor $Q4$ and the diode $D1$. When the output Y is in the low state, $Q2$ and $Q3$ are driven into saturation as in the open-collector gate. The voltage in the collector of $Q2$ is $V_{BE}(Q3) + V_{CE}(Q2)$ or $0.7 + 0.2 = 0.9$ V. The output $Y = V_{CE}(Q3) = 0.2$ V. Transistor $Q4$ is cutoff because its base must be one V_{BE} drop plus one diode drop, or $2 \times 0.6 = 1.2$ V, to start conducting. Since the collector of $Q2$ is connected to the base of $Q4$, the latter's voltage is only 0.9 V instead of the required 1.2 V, and so $Q4$ is cutoff. The reason for placing[1] the diode in the circuit is to provide a diode drop in the output path and thus ensure that $Q4$ is cutoff when $Q3$ is saturated.

When the output changes to the high state because one of the inputs drops to the low state, transistors $Q2$ and $Q3$ go into cutoff. However, the output remains momentarily low because the voltages across the load capacitance cannot change instantaneously. As soon as $Q2$ turns off, $Q4$ conducts because its base is connected to V_{CC} through the 1.6-kΩ resistor. The current needed to charge the load capacitance causes $Q4$ to momentarily saturate, and the output voltage rises with a time constant RC. But R in this case is equal to 130 Ω, plus the saturation resistance of $Q4$, plus the resistance of the diode, for a total of approximately 150 Ω. This value of R is much smaller than the passive pull-up resistance used in the open-collector circuit. As a consequence, the transition from the low to high level is much faster.

As the capacitive load charges, the output voltage rises and the current in $Q4$ decreases, bringing the transistor into the active region. Thus, in contrast to the other transistors, $Q4$ is in the *active* region when in a steady-state condition. The final value of the output voltage is then 5 V, minus a V_{BE} drop in $Q4$, minus a diode drop in $D1$ to about 3.6 V. Transistor $Q3$ goes into cutoff very fast, but during the initial transition time both $Q3$ and $Q4$ are on and a peak current is drawn from the power supply. This current spike generates noise in the power supply distribution system. When the change of state is frequent, the transient current spikes increase the power supply current requirement and the average power dissipation of the circuit increases.

The wired-logic connection is not allowed with totem-pole output circuits. When two totem-poles are wired together with the output of one gate high and the output of the second gate low, the excessive amount of current drawn can produce enough heat to damage the transistors in the circuit (see Problem 13-7). Some TTL gates are constructed to withstand the amount of current that flows under this condition. In any case, the collector current in the low gate may be high enough to move the transistor into the active region and produce an output voltage in the wired connection greater than 0.8 V, which is not a valid binary signal for TTL gates.

13.5.3 Schottky TTL Gate

As mentioned before, a reduction in storage time results in a reduction of propagation delay. This is because the time needed for a transistor to come out of saturation delays the switching of the transistor from the on condition to the off condition. Saturation can be eliminated by placing a Schottky diode between the base and collector of each saturated transistor in the circuit. The Schottky diode is formed by the junction of a metal and semiconductor, in contrast to a conventional diode which is formed by the junction of p-type and n-type semiconductor material. The voltage across a conducting Schottky diode is only 0.4 V, as compared to 0.7 V in a conventional diode. The presence of a Schottky diode between the base and collector prevents the transistor from going into saturation. The resulting transistor is called a *Schottky transistor*. The use of Schottky transistors in a TTL decreases the propagation delay without a sacrifice of power dissipation.

The Schottky TTL gate is shown in Fig. 13-15. Note the special symbol used for the Schottky transistors and diodes. The diagram shows all transistors to be of the Schottky type except $Q4$.

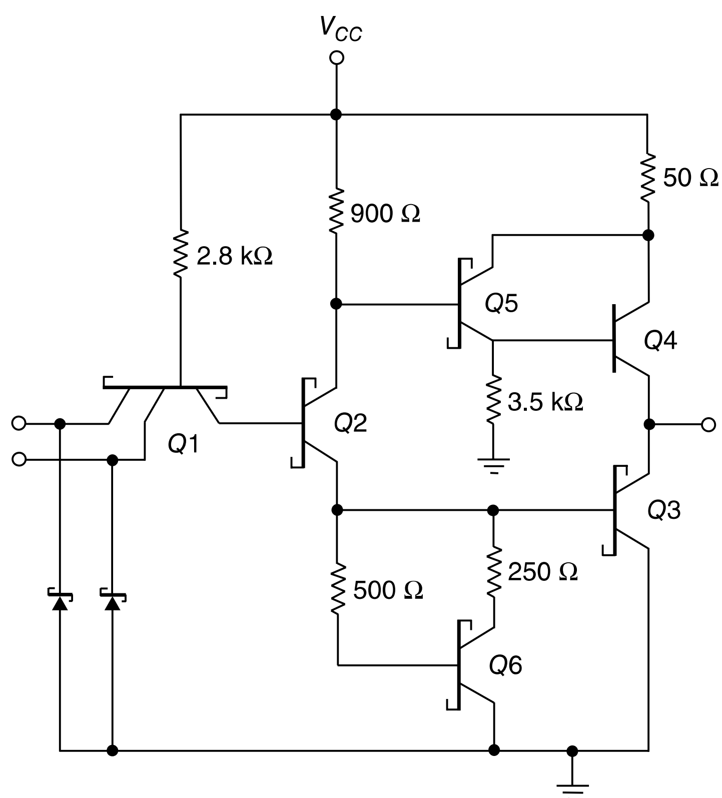


Figure 13.15 Schottky TTL gate

An exception is made of $Q4$ since it does not saturate but stays in the active region. Note also that resistor values have been reduced to further decrease the propagation delay.

In addition to using Schottky transistors and lower resistor values, the circuit of Fig. 13-15 includes other modifications not available in the standard gate of Fig. 13-14. Two new transistors, $Q5$ and $Q6$ have been added, and Schottky diodes are inserted between each input terminal and ground. There is no diode in the totem-pole circuit. However, the new combination of $Q5$ and $Q4$ still gives the two V_{BE} drops necessary to prevent $Q4$ from conducting when the output is low. This combination comprises a double emitter-follower called a *Darlington pair.* The Darlington pair provides a very high current gain and extremely low resistance. This is exactly what is needed during the low-to-high swing of the output, resulting in a decrease of propagation delay.

The diodes in each input shown in the circuit help clamp any ringing that may occur in the input lines. Under transient switching conditions, signal lines appear inductive; this, along with stray capacitance, cause signals to oscillate or "ring." When the output of a gate switches from the high to the low state, the ringing waveform at the input may have excursions below ground as great as 2–3 V, depending on line length. The diodes connected to ground help clamp this ringing since they conduct as soon as the negative voltage exceeds 0.4 V. When the negative excursion is limited, the positive swing is also reduced. The success of the clamp diodes in limiting line effects has been so successful that all versions of TTL gates use them.

The emitter resistor of $Q2$ in Fig. 13-14 has been replaced in Fig. 13-15 by a circuit consisting of transistor $Q6$ and two resistors. The effect of this circuit is to reduce the turn-off current spikes discussed previously. The analysis of this circuit, which helps to reduce the propagation time of the gate, is too involved to present in this brief discussion.

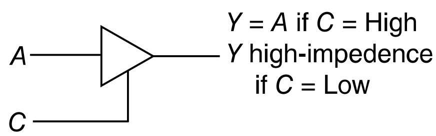
13.5.4 Three-state Gate

As mentioned earlier, the outputs of two TTL gates with totem-pole structures cannot be connected together as in open-collector outputs. There is, however, a special type of totem-pole gate that allows the wired connection of outputs for the purpose of forming a common-bus system. When a totem-pole output TTL gate has this property, it is called a *three-state* (or tri-state) gate.

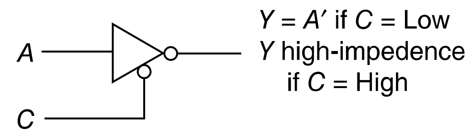
A three-state gate exhibits three output states: (1) a low-level state when the lower transistor in the totem-pole is on and the upper transistor is off; (2) a high-level state when the upper transistor in the totem-pole is on and the lower transistor is off: and (3) a third state when both transistors in the totem-pole are off. The third state provides an open circuit or high-impedance state which allows a direct wire connection of many outputs to a common line. Three-state gates eliminate the need for open-collector gates in bus configurations.

Figure 13-16(a) shows the graphic symbol of a three-state buffer gate. When the control input C is high, the gate is enabled and behaves like a normal buffer with the output equal to the input binary value. When the control input is low, the output is an open circuit which gives a high impedance (the third state) regardless of the value of input A. Some three-state gates produce a high-impedance state when the control input is high. This is shown symbolically in Fig. 13-16(b). Here we have two small circles, one for the inverter output and the other to indicate that the gate is enabled when C is low.

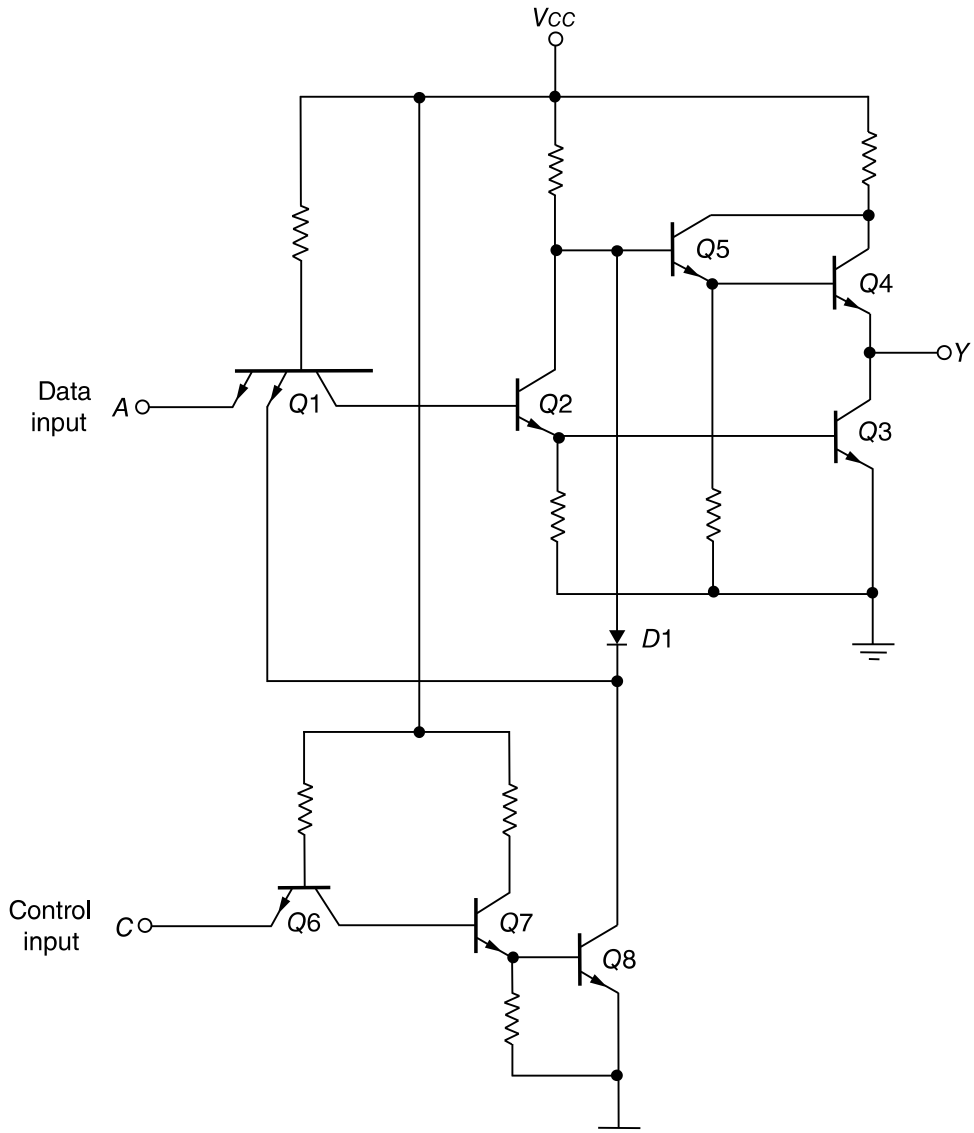
The circuit diagram of the three-state inverter is shown in Fig. 13-16(c). Transistors $Q6$, $Q7$, and $Q8$ associated with the control input form a circuit similar to the open-collector gate. Transistors $Q1$–$Q5$, associated with the data input, form a totem-pole TTL circuit. The two circuits are connected together through diode $D1$. As in an open-collector circuit, transistor $Q8$ turns off



(a) Three-state buffer gate



(b) Three-state inverter gate



(c) Circuit diagram for the three-state inverter of (b)

Figure 13.16 Three-state TTL gate

when the control input at *C* is in the low-level state. This prevents diode *D*1 from conducting, and also, the emitter in *Q*1 connected to *Q*8 has no conduction path. Under this condition, transistor *Q*8 has no effect on the operation of the gate and the output in *Y* depends only on the data input at *A*.

When the control input is high, transistor *Q*8 turns on, and the current flowing from V_{CC} through diode *D*1 causes transistor *Q*8 to saturate. The voltage at the base of *Q*5 is now equal to the voltage across the saturated transistor, $Q8$, plus one diode drop, or 0.9 V. This voltage turns off $Q5$ and $Q4$ since it is less than two V_{BE} drops. At the same time, the low input to one of the emitters of $Q1$ forces transistor $Q3$ (and $Q2$) to turn off. Thus both $Q3$ and $Q4$ in the totem-pole are turned off and the output of the circuit behaves like an open circuit with a very high output impedance.

A three-state bus is created by wiring several three-state outputs together. At any given time, only one control input is enabled while all other outputs are in the high-impedance state. The single gate not in a high-impedance state can transmit binary information through the common bus. Extreme care must be taken that all except one of the outputs are in the third state; otherwise, we have the undesirable condition of having two active totem-pole outputs connected together.

An important feature of most three-state gates is that the output enable delay is longer than the output disable delay. If a control circuit enables one gate and disables another at the same time, the disabled gate enters the high-impedance state before the other gate is enabled. This eliminates the situation of both gates being active at the same time.

There is a very small leakage current associated with the high-impedance condition in a three-state gate. Nevertheless, this current is so small that as many as 100 three-state outputs can be connected together to form a common bus line.

13.6 Emitter-coupled Logic (ECL)

Emitter-coupled logic (ECL) is a nonsaturated digital logic family. Since transistors do not saturate, it is possible to achieve propagation delays of 2 ns and even below 1ns. This logic family has the lowest propagation delay of any family and is used mostly in systems requiring very-high-speed operation. Its noise immunity and power dissipation, however, are the worst of all the logic families available.

A typical basic circuit of the ECL family is shown in Fig. 13-17. The outputs provide both the OR and NOR functions. Each input is connected to the base of a transistor. The two voltage levels are about −0.8 V for the high state and about −1.8 V for the low state. The circuit consists of a differential amplifier, a temperature- and voltage-compensated bias network, and an emitter-follower output. The emitter outputs require a pull-down resistor for current to flow. This is obtained from the input resistor, R_P, of another similar gate or from an external resistor connected to a negative voltage supply.

The internal temperature- and voltage-compensated bias circuit supplies a reference voltage to the differential amplifier. Bias voltage V_{BB} is set at −1.3 V, which is the midpoint of the signal logic swing. The diodes in the voltage divider, together with $Q6$, provide a circuit that maintains a constant V_{BB} value despite changes in temperature or supply voltage. Any one of the power supply inputs could be used as ground. However, the use of the V_{CC} node as ground and V_{EE} at $-5.2\ V$ results in best noise immunity.

If any input in the ECL gate is high, the corresponding transistor is turned on and $Q5$ is turned off. An input of − 0.8 V causes the transistor to conduct and places −1.6 V on the emitters of all transistors (V_{BE} drop in ECL transistors is 0.8 V). Since $V_{BB} = -1.3$ V. the base voltage of $Q5$ is only $0.3\ V$ more positive than its emitter. $Q5$ is cutoff because its V_{BE} voltage needs at least 0.6 V to start conducting. The current in resistor R_{C2} flows into the base of $Q8$ (provided there is a load resistor). This current is so small that only a negligible voltage drop occurs across R_{C2}. The OR output of the gate is one V_{BE} drop below ground, or −0.8 V, which is the high state. The current flowing through R_{C1} and the conducting transistor causes a drop of about 1 V below

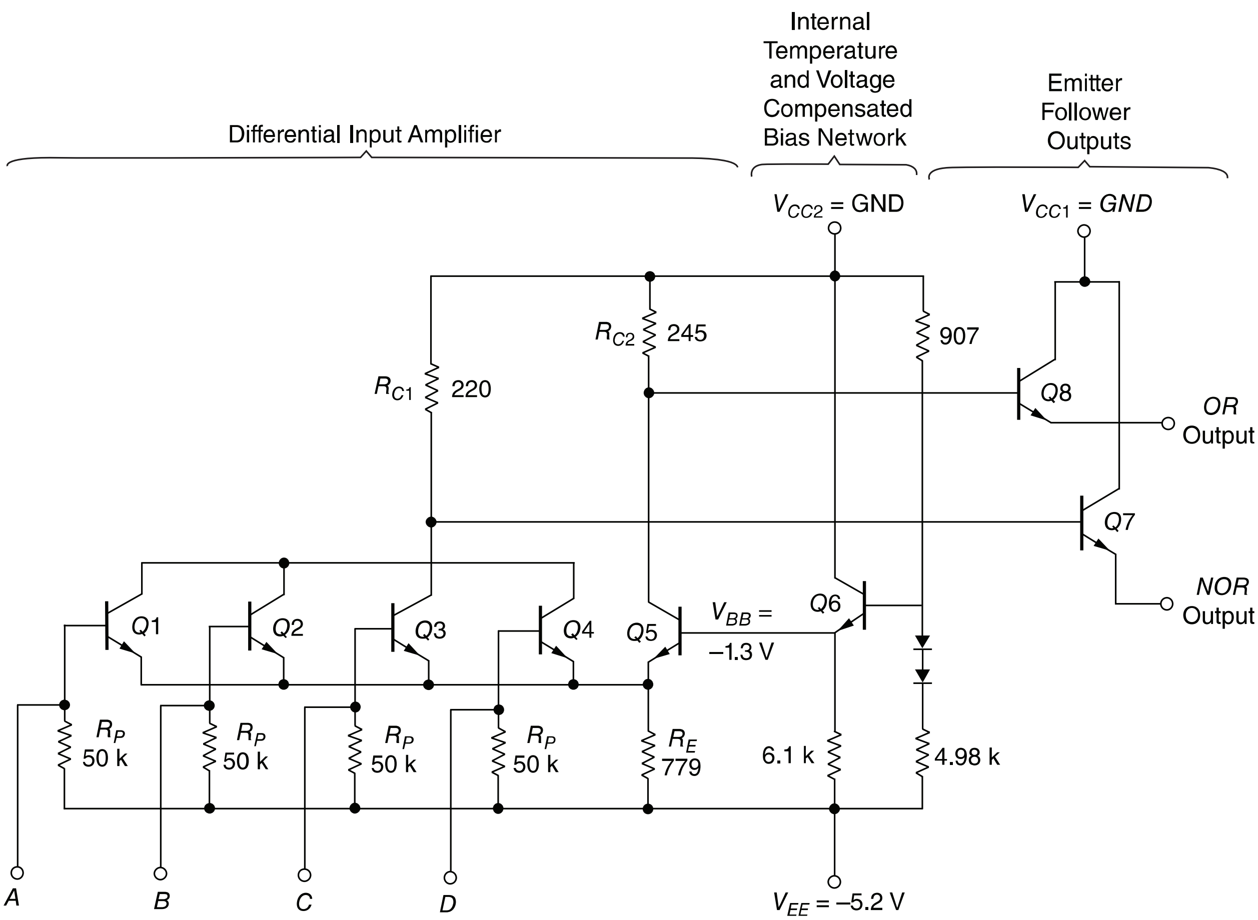


Figure 13.17 ECL basic gate

ground (see Problem 13-9). The NOR output is one V_{BE} drop below this level, or at −1.8 V, which is the low state.

If all inputs are at the low level, all input transistors turn off and $Q5$ conducts. The voltage in the common-emitter node is one V_{BE} drop below V_{BB} or − 2.1 V. Since the base of each input is at a low level of −1.8 V, each base-emitter junction has only 0.3 V and all input transistors are cutoff. R_{C2} draws current through $Q5$ that results in a voltage drop of about 1 V, making the OR output one V_{BE} drop below this, at −1.8 V or the low level. The current in R_{C1} is negligible and the NOR output is one V_{BE} drop below ground, at −0.8 V or the high level. This verifies the OR and NOR operations of the circuit.

The propagation delay of the ECL gate is 2 ns, and the power dissipation is 25 mW. This gives a speed-power product of 50, which is about the same as for Schottky TTL. The noise margin is about 0.3 V and not as good as in the TTL gate. High fan-out is possible in the ECL gate because of the high input impedance of the differential amplifier and the low output impedance of the emitter-follower. Because of the extreme high speed of the signals, external wires act like transmission lines. Except for very short wires of a few centimeters, ECL outputs must use coaxial cables with a resistor termination to reduce line reflections.

The graphic symbol for the ECL gate is shown in Fig. 13-18(a). Two outputs are available, one for the NOR function and the other for the OR function. The outputs of two or more ECL gates can be connected together to form wired logic. As shown in Fig. 13-18(b), an *external*

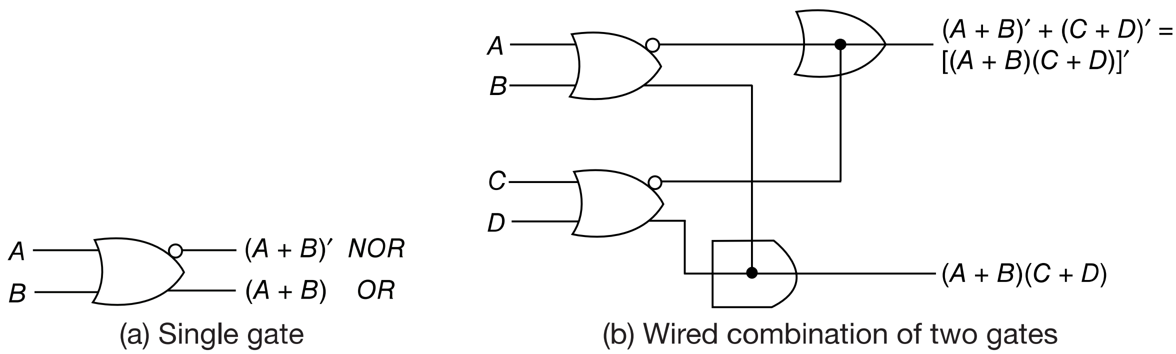


Figure 13.18 Graphic symbols of ECL gates

wired connection of two NOR outputs produces a wired-OR function. An *internal* wired connection of two OR outputs is employed in some ECL ICs to produce a wired-AND (sometimes called dot-AND) logic. This property may be utilized when ECL gates are used to form the OR-AND-INVERT and the OR-AND functions.

13.7 Metal-Oxide Semiconductor (MOS)

The field-effect transistor (FET) is a unipolar transistor, since its operation depends on the flow of only one type of carrier. There are two types of field-effect transistors: the junction field-effect transistor (JFET) and the metal-oxide semi-conductor (MOS). The former is used in linear circuits and the latter in digital circuits. MOS transistors can be fabricated in less area than bipolar transistors.

The basic structure of the MOS transistor is shown in Fig. 13-19. The p-channel MOS consists of a lightly doped substrate of n-type silicon material. Two regions are heavily doped by diffusion with p-type impurities to form the *source* and *drain.* The region between the two p-type sections serves as the *channel.* The *gate* is a metal plate separated from the channel by an insulated dielectric of silicon dioxide. A negative voltage (with respect to the substrate) at the gate terminal causes an induced electric field in the channel which attracts p-type carriers from the substrate. As the magnitude of the negative voltage on the gate increases, the region below the gate accumulates more positive carriers, the conductivity increases, and current can flow from source to drain provided a voltage difference is maintained between these two terminals.

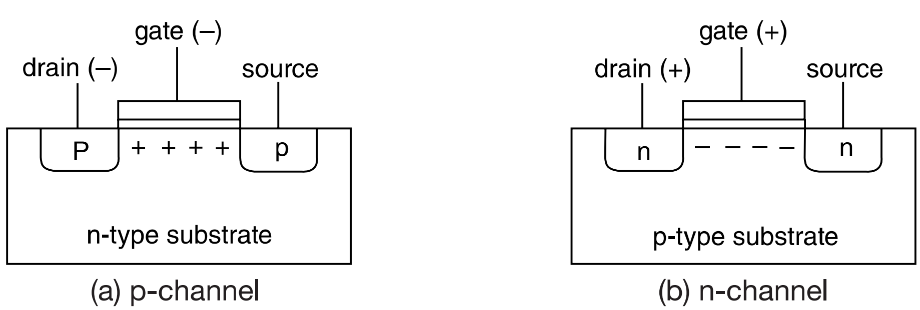


Figure 13.19 Basic structure of MOS transistor

There are four basic types of MOS structures. The channel can be a p- or n-type, depending on whether the majority carriers are holes or electrons. The mode of operation can be enhancement or depletion, depending on the state of the channel region at zero gate voltage. If the channel is initially doped lightly with p-type impurity (diffused channel), a conducting channel exists at zero gate voltage and the device is said to operate in the *depletion* mode. In this mode, current flows unless the channel is depleted by an applied gate field. If the region beneath the gate is left initially uncharged, a channel must be induced by the gate field before current can flow. Thus, the channel current is enhanced by the gate voltage and such a device is said to operate in the *enhancement* mode.

The source is the terminal through which the majority carriers enter the bar. The drain is the terminal through which the majority carriers leave the bar. In a p-channel MOS, the source terminal is connected to the substrate and a negative voltage is applied to the drain terminal. When the gate voltage is above a threshold voltage V_T (about −2 V), no current flows in the channel and the drain-to-source path is like an open circuit. When the gate voltage is sufficiently negative below V_T, a channel is formed and p-type carriers flow from source to drain. P-type carriers are positive and correspond to a positive current flow from source to drain.

In the n-channel MOS, the source terminal is connected to the substrate and a positive voltage is applied to the drain terminal. When the gate voltage is below the threshold voltage V_T (about 2 V) no current flows in the channel, When the gate voltage is sufficiently positive above V_T to form the channel, n-type carriers flow from source to drain. N-type carriers are negative, which corresponds to a positive current flow from drain to source. The threshold voltage may vary from 1 to 4 V depending on the particular process used.

The graphic symbols for the MOS transistors are shown in Fig. 13-20. The accepted symbol for the enhancement type is the one with the broken-line connection between source and drain. In this symbol, the substrate can be identified and is shown connected to the source. We will use an alternative symbol that omits the substrate; in this symbol, the arrow is placed in the source terminal to show the direction of *positive* current flow (from source to drain in the p-channel and from drain to source in the n-channel).

Because of the symmetrical construction of source and drain, the MOS transistor can be operated as a bilateral device. Although normally operated so that carriers flow from source to drain, there are circumstances when it is convenient to allow carrier flow from drain to source (see Problem 13-12).

One advantage of the MOS device is that it can be used not only as a transistor, but as a resistor as well. A resistor is obtained from the MOS by permanently biasing the gate terminal for conduction. The ratio of the source-drain voltage to the channel current then determines the value of the resistance. Different resistor values may be constructed during manufacturing by fixing the channel length and width of the MOS device.

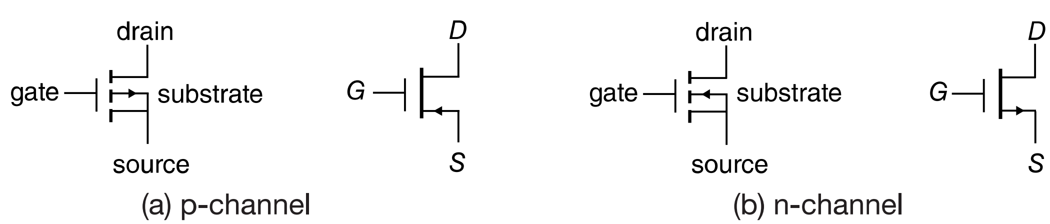


Figure 13.20 Symbols for MOS transistors

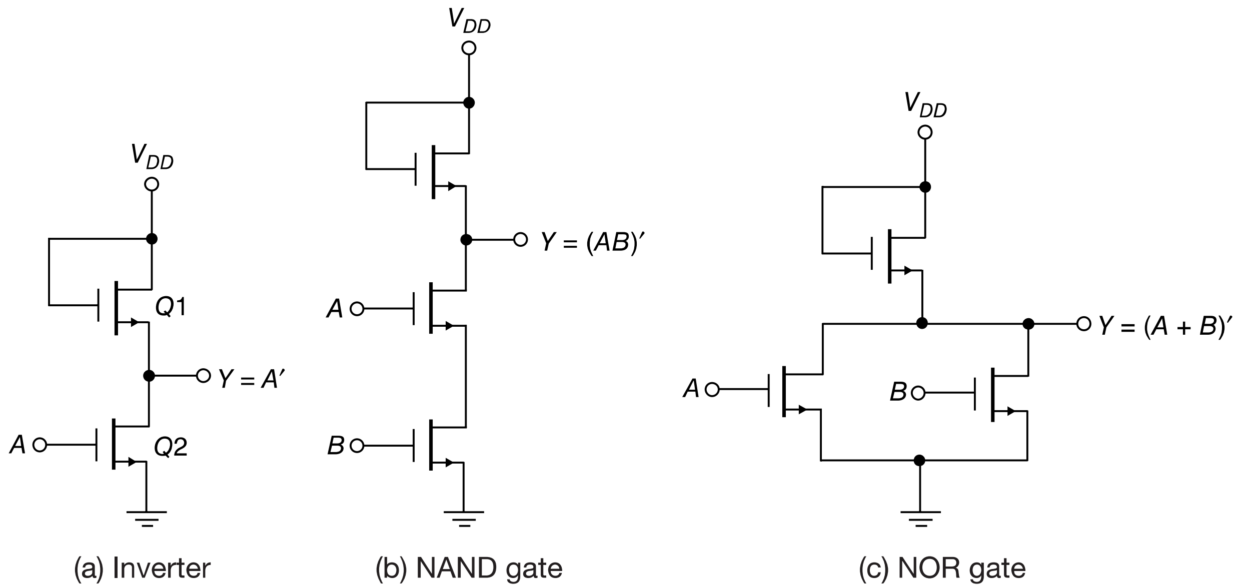


Figure 13.21 n-channel MOS logic circuits

Three logic circuits using MOS devices are shown in figure 13-21. For an n-channel MOS, supply voltage V_{DD} is positive (about 5 V) to allow positive current flow from drain to source. The two voltage levels are a function of the threshold voltage V_T. The low level is anywhere from zero to V_T, and the high level ranges V_T to V_{DD}. The n-channel gates usually employ positive logic. The p-channel MOS circuits use a negative voltage for V_{DD} to allow positive current flow from source to drain. The two voltage levels are both negative above and below the negative threshold voltage V_T. P-channel gates usually employ negative logic.

The inverter circuit shown in Fig. 13-21(a) uses two MOS devices. *Q*1 acts as the load resistor and *Q*2 as the active device. The load resistor MOS has its gate connected to V_{DD}, thus maintaining it always in the conduction state. When the input voltage is low (below V_T), *Q*2 turns off. Since *Q*1 is always on, the output voltage is at about V_{DD}. When the input voltage is high (above V_T), *Q*2 turns on. Current flows from V_{DD} through the load resistor *Q*1 and into *Q*2. The geometry of the two MOS devices must be such that the resistance of *Q*2, when conducting, is much less than the resistance of *Q*1 to maintain the output *Y* at a voltage below V_T.

The NAND gate shown in Fig. 13-21(b) uses transistors in series. Inputs *A* and *B* must both be high for all transistors to conduct and cause the output to go low. If either input is low, the corresponding transistor is turned off and the output is high. Again, the series resistance formed by the two active MOS devices must be much less than the resistance of the load resistor MOS. The NOR gate shown in Fig. 13-21(c) uses transistors in parallel. If either input is high, the corresponding transistor conducts and the output is low. If all inputs are low, all active transistors are off and the output is high.

13.8 Complementary MOS (CMOS)

Complementary MOS circuits take advantage of the fact that both n-channel and p-channel devices can be fabricated on the same substrate. CMOS circuits consist of both types of MOS

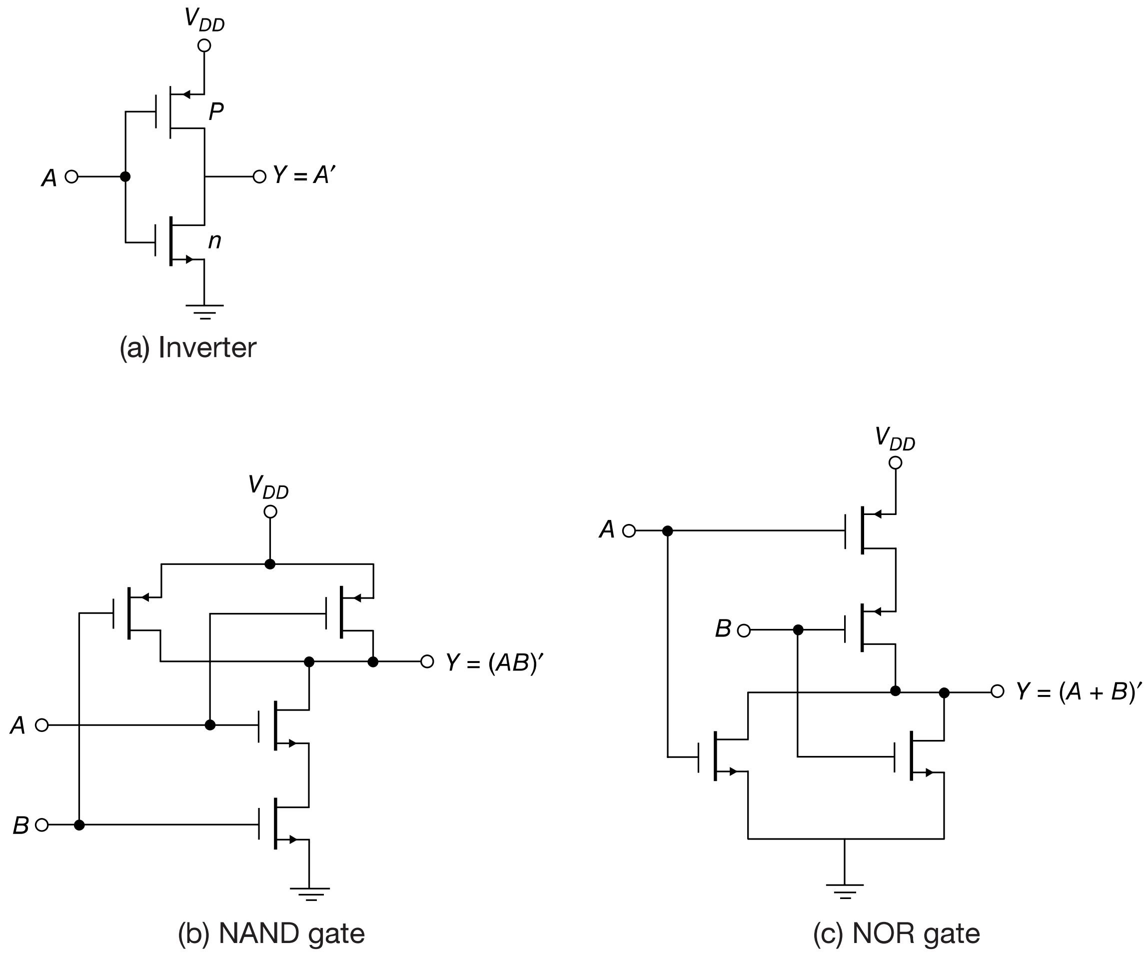


Figure 13.22 CMOS logic circuits

devices interconnected to form logic functions. The basic circuit is the inverter, which consists of one p-channel transistor and one n-channel transistor as shown in Fig. 13-22(a). The source terminal of the p-channel device is at V_{DD}, and the source terminal of the n-channel device is at ground. The value of V_{DD} may be anywhere from +3 to +18 V. The two voltage levels are 0 V for the low level and V_{DD} for the high level.

To understand the operation of the inverter, we must review the behavior of the MOS transistor from the previous section:

1. The n-channel MOS conducts when its gate-to-source voltage is positive.
2. The p-channel MOS conducts when its gate-to-source voltage is negative.
3. Either type of device is turned off if its gate-to-source voltage is zero.

Now consider the operation of the inverter. When the input is low, both gates are at zero potential. The input is at $-V_{DD}$ relative to the source of the p-channel device and at 0 V relative to the source of the n-channel device. The result is that the p-channel device is turned on and the n-channel device is turned off. Under these conditions, there is a low-impedance path from V_{DD} to the output and a very-high-impedance path from output to ground. Therefore, the output voltage approaches the high level V_{DD} under normal loading conditions. When the input is high, both gates are at V_{DD} and the situation is reversed: The p-channel device is off and the n-channel device is on. The result is that the output approaches the low level of 0 V.

In either logic state, one MOS transistor is on while the other is off. Because one transistor is always turned off, the dc power dissipation of the CMOS circuit is extremely low, usually on the order of 10 nW. The major power drain occurs when the CMOS circuit changes state.

CMOS logic is usually specified for single-supply operation over the 5–15V range, but some circuits may be operated at 3 V or 18 V, Operating CMOS at large values of supply voltage produces a greater power dissipation. The propagation delay time decreases and the noise margin improves with increased power supply voltage. The propagation delay of the inverter is about 25 ns. The noise margin is usually about 40% of the V_{DD} supply voltage value. The advantages of CMOS, i.e., low power dissipation, excellent noise immunity, high packing density, and a wide range of supply voltages, make it a strong contender for a popular standard as a digital circuit family.

Two other CMOS basic gates are shown in Fig. 13-22. A two-input NAND gate consists of two p-type units in parallel and two n-type units in series, as shown in Fig. 13-22(b). If all inputs are high, both p-channel transistors turn off and both n-channel transistors turn on. The output has a low impedance to ground and produces a low state. If any input is low, the associated n-channel transistor is turned off and the associated p-channel transistor is turned on. The output is coupled to V_{DD} and goes to the high state. Multiple-input NAND gates may be formed by placing equal numbers of p-type and n-type transistors in parallel and series, respectively, in an arrangement similar to that shown in Fig. 13-22(b).

A two-input NOR gate consists of two n-type units in parallel and two p-type units in series, as shown in Fig. 13-22(c). When all inputs are low, both p-channel units are on and both n-channel units are off. The output is coupled to V_{DD} and goes to the high state. If any input is high, the associated p-channel transistor is turned off and the associated n-channel transistor turns on. This connects the output to ground, causing a low-level output.

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PROBLEMS

13-1. (a) Determine the high-level output voltage of the RTL gate for a fan-out of 5. (b) Determine the minimum input voltage required to drive an RTL transistor to saturation when $h_{FE} = 20$. (c) From the results in (a) and (b), determine the noise margin of the RTL gate when the input is high and the fan-out is 5.

13-2. Show that the output transistor of the DTL gate of Fig. 13-5 goes into saturation when all inputs are high. Assume that $h_{FE} = 20$.

13-3. Connect the output Y of the DTL gate shown in Fig. 13-5 to N inputs of other similar gates. Assume that the output transistor is saturated and its base current is 0.44 mA. Let $h_{FE} = 20$.

(a) Calculate the current in the 2-kΩ resistor.

(b) Calculate the current coming from each input connected to the gate.

(c) Calculate the total collector current in the output transistor as a function of N.

(d) Find the value of N that will keep the transistor in saturation.

(e) What is the fan-out of the gate?

13-4. Draw the interconnection of I^2L gates to form a 2 × 4 decoder.

13-5. Let all inputs in the open-collector TTL gate of Fig. 13-11 be in the high state of 3 V.

(a) Determine the voltages in the base, collector, and emitter of all transistors.

(b) Determine the minimum h_{FE} of $Q2$ that ensures that this transistor saturates.

(c) Calculate the base current of $Q3$.

(d) Assume that the minimum h_{FE} of $Q3$ is 6.18. What is the maximum current that can be tolerated in the collector to ensure saturation of $Q3$?

(e) What is the minimum value of R_L that can be tolerated to ensure saturation of $Q3$?

13-6. (a) Using the actual output transistors of two open-collector TTL gates, show (by means of a truth table) that when connected together to an external resistor and V_{CC}, the wired connection produces an AND function. (b) Prove that two open-collector TTL inverters when connected together produce the NOR function.

13-7. It was stated in Section 13-5 that totem-pole outputs should not be tied together to form wired logic. To see why this is prohibitive, connect two such circuits together and let the output of one gate be in the high state and the output of the other gate be in the low state. Show that the load current (which is the sum of the base and collector currents of the saturated transistor $Q4$ in Fig. 13-14) is about 32 mA. Compare this value with the recommended load current in the high state of 0.4 mA.

13-8. For the following conditions, list the transistors that are off and those that are conducting in the three-state TTL gate of Fig. 13-16(c). (For $Q1$ and $Q6$, it would be necessary to list the states in the base-emitter and base-collector junctions separately.)

(a) When C is low and A is low.

(b) When C is low and A is high.

(c) When C is high.

What is the state of the output in each case?

13-9. Calculate the emitter current I_E across R_E in (he ECL gate of Fig. 13-17 when:

(a) At least one input is high at −0.8 V.

(b) All inputs are low at −1.8 V.

Now assume that $I_C = I_E$. Calculate the voltage drop across the collector resistor in each case and show that it is about 1 V as required.

13-10. Calculate the noise margin of the ECL gate.

13-11. Using the NOR outputs of two ECL gates, show that when connected together to an external resistor and negative supply voltage, the wired connection produces an OR function.

13-12. The MOS transistor is bilateral, i. e., current may flow from source to drain or from drain to source. Using this property, derive a circuit that implements the Boolean function:

$$Y = (AB + CD + AED + CEB)'$$

using six MOS transistors.

13-13. (a) Show the circuit of a four-input NAND gate using CMOS transistors. (b) Repeat for a four-input NOR gate.

MORE SOLVED QUESTIONS

1. Why we go for Digital integrated circuits?

   Digital : Processing of discretized information (signal).

   1. Accurate.
   2. Flexible.
   3. It requires less operational energy.
   4. Repeatability.

2. Why there are different Logic Families?

   1. Space satellite, Digital wristwatch – low power consumption
   2. Scientific Computer – high speed
   3. Digital Control of equipment in industrial environment – noise immunity
   4. Compatibility
   5. Flexibility
   6. Cost
   7. Size, Packing Density
   8. Evolution

3. Transistor-inverter is key to development of logic circuit. Why?

   Diodes, Switches can be used to build OR and AND circuit but not NOT circuit. Hence, with these it is not possible to develop every type of logic circuits.

Appendix

ANSWERS TO SELECTED PROBLEMS

Chapter 13

13-1. (a) 1.05 V
(b) 0.82 V
(c) 0.23 V

13-2. $I_B = 0.44$ mA, $I_{cs} = 2.4$ mA

13-3. (a) 2.4 mA
(b) 0.82 mA
(c) 2.4 + 0.82 N
(d) 7.8
(e) 7

13-5. (b) 3.53
(c) 2.585 mA
(d) 16 mA
(e) 300 Ω

13-9. (a) 4.62 mA
(b) 4 mA

13-10. 0.3 V.